

ITA0608 - MACHINE LEARNING

LAB PROGRAMS

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Experiment 10: Expectation-Maximization (EM) Algorithm

Aim

To implement the Expectation-Maximization (EM) algorithm in Python for clustering using an appropriate dataset.

Procedure

1. Import the required libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Standardize the input features.
5. Create the Gaussian Mixture Model (GMM).
6. Train the model using the Expectation-Maximization algorithm.
7. Predict the cluster labels.
8. Display the cluster assignments and model parameters.

Algorithm

1. Start.
2. Load the dataset.
3. Normalize the input features.
4. Initialize the Gaussian Mixture Model.
5. Perform the Expectation (E) Step to calculate the probability of each data point belonging to each cluster.
6. Perform the Maximization (M) Step to update the model parameters.
7. Repeat the E-Step and M-Step until convergence.
8. Predict the cluster labels.
9. Display the clustering results.
10. Stop.

Code

```
from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.mixture import GaussianMixture

data = load_iris()

X = data.data

scaler = StandardScaler()

X = scaler.fit_transform(X)

model = GaussianMixture(
    n_components=3,
    random_state=42
)

model.fit(X)

labels = model.predict(X)

print("Cluster Labels:")
print(labels)

print("Means:")
print(model.means_)

print("Weights:")
print(model.weights_)
```

Output

```
... Cluster Labels:  
[1 1 1 1 1 1 1 2 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1  
 1 2 1 1 2 2 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 2 0 0 0 0 0 0 0 0 0 0  
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0  
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0  
 0 0]  
  
Means:  
[[ 0.53745909 -0.39369142  0.6693573   0.64500292]  
 [-0.93852253  0.98617415 -1.29410958 -1.24871335]  
 [-1.53616188 -0.9148767  -1.05760659 -1.00758605]]  
  
Weights:  
[0.65376794 0.30201321 0.04421885]
```

Result

The Expectation-Maximization (EM) algorithm was successfully implemented in Python using the Gaussian Mixture Model (GMM). The model clustered the dataset by iteratively performing the Expectation and Maximization steps until convergence.